

IPA ASSIGNMENT – 17

Name – Venya Kejriwal

Roll No – 24155517

1. Store one RGB image using 3 2d matrices

```
from PIL import Image
import numpy as np
image = Image.open("img_2.jpeg").convert("RGB")
img_array = np.array(image)
R = img_array[:, :, 0]
G = img_array[:, :, 1]
B = img_array[:, :, 2]
print("Red Matrix:")
print(R)
print("\nGreen Matrix:")
print(G)
print("\nBlue Matrix:")
print(B)
print("\nImage Size:", image.size)
```

OUTPUT

```
Red Matrix:
[[ 0  0  0  ... 178 175 176]
 [ 0  0  0  ... 179 175 176]
 [ 0  0  0  ... 179 176 175]
 ...
 [ 37 38 38  ... 115 111 102]
 [ 37 38 38  ... 115 121 120]
 [ 37 38 38  ... 129 146 157]]

Green Matrix:
[[123 123 123  ... 170 172 172]
 [123 123 123  ... 171 172 172]
 [122 122 122  ... 171 173 171]
 ...
 [ 31 32 32  ...  82  76  68]
 [ 32 33 33  ...  80  85  83]
 [ 32 33 33  ...  93 109 117]]

Blue Matrix:
[[159 159 159  ... 167 167 171]
 [159 159 159  ... 168 167 169]
 [159 159 159  ... 168 168 168]
 ...
 [ 57 58 58  ...  28  18  5]
 [ 55 56 56  ...  38  37  30]
 [ 55 56 56  ...  57  65  68]]

Image Size: (1289, 860)
```

2. Implement rgb to grayscale of the image

```
import cv2
import matplotlib.pyplot as plt
image = cv2.imread("img_2.jpeg")
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original RGB Image")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(gray_image, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
cv2.imwrite("grayscale_image.jpg", gray_image)
```

OUTPUT

